

$$\begin{array}{ccc}
FX & \xleftarrow{i_{FX}} & [\mathbb{1}_{\mathcal{D}}, FX]_{\mathcal{D}} \\
\uparrow Fi_X & & \uparrow [\varphi_0, FX]_{\mathcal{D}} \\
F([\mathbb{1}_c, X]c) & \xrightarrow{\varphi_{\mathbb{1}_c, X}} & [F\mathbb{1}_c, FX]_{\mathcal{D}}
\end{array}$$